

Regular expressions (regex)

Thursday, June 18, 2026

i Today: Thursday June 18

Before class

- **Perusall Reading** (Canvas): [Data Computing Ch. 17](#) §17.1–17.2 — Regular expressions

Plan for today

- **Project oral check-in #1**; work on the activity if your group doesn't go first.
- What a **regex** and its roles (**detect**, **replace**, **extract**)
- literals, `.`, character classes `[ ]`, alternation `|`
- **Quantifiers** `?` `*` `+` `{n}`, **anchors** `^` `$`, and **escaping**
- **Capturing** with `( )` and pulling pieces out
- In-class activity

Helpful links

- [Syllabus](#) · [AI policy](#) · [Schedule](#) · [DC Ch. 17](#) · [stringr cheatsheet](#) · [regex101.com](#)

What is a regex?

Patterns in strings

A **regular expression** (regex) is a compact way to describe a **pattern** in text. Lots of everyday strings follow patterns:

Kind	Examples
Phone numbers	(651) 696-6000 · +1 651.696.6000

Kind	Examples
Times	10:30 AM · 13:15.54
Durations	1:59:40 (Kipchoge's sub-2-hour marathon)
US postal codes	16801 · 55105-1362
Stray whitespace	" henry "

A regex lets you say “match anything that has this particular shape” without having to list out every possible case (which is often impossible).

Three jobs regex does

Across all of data cleaning, regex does three things:

Job	base R	stringr (tidyverse)	Often with
Detect a pattern	<code>grepl()</code>	<code>str_detect()</code>	<code>filter()</code>
Replace a pattern	<code>gsub()</code>	<code>str_replace_all()</code>	<code>mutate()</code>
Extract a match	<code>regmatches()</code>	<code>str_extract()</code>	<code>mutate()</code>

The textbook uses base R (`grepl`, `gsub`); the analogous versions from the `stringr` package do the same things but have tidier names.

`grepl()`: does the pattern appear?

`grepl(pattern, x)` returns TRUE/FALSE for each string. The **pattern is the quoted string**. Here: names containing “shine” (`ignore.case = TRUE` ignores capitalization):

```
NameList |>
  filter(grepl("shine", name, ignore.case = TRUE)) |>
  head(5)
```

```
# A tibble: 5 x 3
  name      sex  total
<chr>    <chr> <int>
1 Sunshine F      4959
2 Shineka  F       150
3 Shine    F        95
4 Shine    M        44
5 Sunshine M        37
```

`str_detect(name, regex("shine", ignore_case = TRUE))` does the same thing.

Regex building blocks

Literals & special characters

The simplest patterns are **literal**: "able" matches any string containing "able" ("capable", "tabled") but is **case-sensitive** by default ("Able" won't match unless `ignore.case = TRUE`).

These characters are **special** in that they mean something other than themselves:

. ^ \$ [] { } | () + * ?

For example . means "any single character", so ".ble" matches "able", "soluble", "fungible", "Bible".

Literals & special characters

Character	Meaning	Example
.	Any single character (except a newline)	".ble" matches "able", "soluble", or "Bible".
^	Start of the string	"^a" matches "apple" but not "banana".
\$	End of the string	"e\$" matches "apple" but not "elephant".
[]	Character class (matches <i>any one</i> character inside)	"[ab]c" matches "ac" or "bc".
	OR (alternation)	"cat dog" matches "cat" or "dog".
()	Grouping (treats multiple characters as one unit)	"(ab)+" matches "abab".
*	Zero or more of the preceding item	"a*b" matches "b", "ab", or "aab".
+	One or more of the preceding item	"a+b" matches "ab" or "aab" (but not "b").
?	Zero or one of the preceding item (makes it optional)	"colou?r" matches "color" or "colour".
{ }	Quantifier (specific number or range of repetitions)	"a{2,3}" matches "aa" or "aaa".

Note: If you ever need to match one of these special characters literally (like an actual period or question mark), you have to escape it by putting a backslash in front of it (e.g., `\.` or `\?`). In R, because of how strings work, you usually need *two* backslashes (e.g., `\\.).`

Character classes []

Square brackets mean “**any one** of these characters”:

- `[aeiou]` → any vowel
- `[^aeiou]` → **not** a vowel (the `^` *inside* brackets means “except”) → a consonant
- `[0-9]` → any digit; `[a-z]` → any lowercase letter

```
NameList |>
  filter(grepl("mn", name, ignore.case = TRUE)) |> # literal "mn"
  head(3)
```

```
# A tibble: 3 x 3
  name    sex    total
<chr> <chr> <int>
1 Autumn F      104408
2 Sumner M        2287
3 Amna   F        1099
```

Tip

A vertical bar `|` means “**or**”: `"rain|snow|sleet|hail"` matches any of those four words.

Putting characters in a row

Two patterns side by side means “this **then** that”. `^[^aeiou].[^aeiou].[^aeiou]` reads as:

- `^` start of string, then
- `[^aeiou]` a consonant, `.` anything, `[^aeiou]` a consonant, `.` anything, `[^aeiou]` a consonant

i.e. the 1st, 3rd, and 5th letters are consonants:

```
NameList |>
  filter(grepl("^[^aeiou].[^aeiou].[^aeiou]", name, ignore.case = TRUE)) |>
  head(3)
```

```
# A tibble: 3 x 3
  name    sex    total
  <chr>  <chr>   <int>
1 James  M       5091189
2 Robert M       4789776
3 David  M       3565229
```

Quantifiers

How many times?

A quantifier follows a pattern and says **how many times** it repeats:

Quantifier	Means
?	zero or one time
*	zero or more times
+	one or more times
{n}	exactly n times
{n,m}	between n and m times
{n,}	n or more times

Warning

The *Data Computing* reading swaps the definitions of + and *. **In R (and every real regex engine):** * = zero-or-more, + = one-or-more.

Quantifiers in action

```
x <- c("Mr.", "Ms", "Mrs.", "Miss", "Man", "Miocene")
x[grepl("M[aeiou]+", x)] # M then ONE or more vowels
```

```
[1] "Miss"    "Man"     "Miocene"
```

```
x[grepl("M[aeiou]*", x)] # M then ZERO or more vowels (matches Mr.)
```

```
[1] "Mr."     "Ms"      "Mrs."    "Miss"    "Man"     "Miocene"
```

Three or more vowels in a row, `[aeiou]{3,}`:

```
NameList |>
  filter(grepl("[aeiou]{3,}", name, ignore.case = TRUE)) |>
  head(3)
```

```
# A tibble: 3 x 3
  name    sex    total
  <chr>  <chr>  <int>
1 Louis  M      389910
2 Louise F      331551
3 Isaiah M      177412
```

Anchors & escaping

Start ^ and end \$

Outside of square brackets, ^ and \$ are **anchors**:

- ^ = the **start** of the string
- \$ = the **end** of the string

Names that **end** in a vowel, `[aeiou]$`:

```
NameList |>
  filter(grepl("[aeiou]$", name)) |>
  group_by(sex) |>
  summarise(total = sum(total), .groups = "drop")
```

```
# A tibble: 2 x 2
  sex    total
  <chr>  <int>
1 F      96702371
2 M     21054791
```

Girls' names end in a vowel almost five times as often as boys' names.

Escaping special characters

To match a special character **literally**, **escape** it with `\\`. So `\\.` means a real period, and `\\$` a real dollar sign (not the end-of-string anchor).

```
currency <- c("$100.95", "45¢")
gsub("^\\$|¢$", "", currency) # strip a leading $ or trailing ¢
```

```
[1] "100.95" "45"
```

Here `^\\$` is a literal `$` at the start; `¢$` is a literal `¢` at the end. Without the `\\`, that first `$` would mean “end of string”.

Capturing & extracting

Parentheses capture

Wrap part of a pattern in `( )` to **capture** it; in other words, pull that piece out. `str_extract()` returns the matched text; with capture groups, `gsub()` can refer back to group 1 as `\\1`.

Names with **four vowels in a row**, extracting those vowels:

```
BabyNames |>
  group_by(name) |>
  summarise(total = sum(count), .groups = "drop") |>
  mutate(match = str_extract(name, "([aeiouAEIOU]{4})") |>
  filter(!is.na(match)) |>
  arrange(desc(total)) |>
  head(4)
```

```
# A tibble: 4 x 3
  name      total match
  <chr>    <int> <chr>
1 Louie   29293 ouie
2 Sequoia 3275  uoia
3 Gioia    480  ioia
4 Keiaira   81  eiai
```

Backreferences with \\1

The regex `".*([aeiou])$"` = “anything, then a captured final vowel”. In `gsub()`, `replacement = "\\1"` keeps **only** the captured group, so the whole name collapses to its last vowel:

```
re <- ".*([aeiou])$"
NameList |>
  filter(grepl(re, name)) |>
  mutate(vowel = gsub(re, "\\1", name)) |>
  group_by(sex, vowel) |>
  summarise(total = sum(total), .groups = "drop") |>
  pivot_wider(names_from = sex, values_from = total)
```

```
# A tibble: 5 x 3
  vowel      F      M
  <chr>    <int>  <int>
1 a      56088501 1844041
2 e      36432218 14341114
3 i       3693024  753311
4 o       403120  4041190
5 u       85508   75135
```

Names ending in “a”, “e”, “i” skew female; “o” skews male.

Cleaning with regex

Stripping junk out of numbers

Scraped numbers arrive with \$, , %, units. A character class lists everything to delete at once:

```
debt <- tibble(
  country = c("World", "United States*", "Japan"),
  debt     = c("$56,308 billion", "$17,607 billion", "$9,872 billion"),
  percGDP  = c("64%", "73.60%", "214.30%")
)
debt |>
  mutate(
    debt     = gsub("[$,%]|billion", "", debt),
    percGDP  = gsub("[,%]", "", percGDP)
  )
```

```
# A tibble: 3 x 3
  country      debt    percGDP
  <chr>        <chr>    <chr>
1 World      "56308 " 64
2 United States* "17607 " 73.60
3 Japan      "9872 " 214.30
```

[`$`,`%`] deletes any `$`, `,`, or `%`; `|billion` also deletes the word “billion”.

Trimming whitespace

`^ +| +$` matches one-or-more spaces at the **start** *or* the **end**:

```
s <- "  My name is Julia  "
gsub("^ +| +$", "", s)
```

```
[1] "My name is Julia"
```

Tip

`stringr` has friendly shortcuts for these: `str_remove_all()`, `str_trim()` (ends), and `str_squish()` (ends + collapse internal runs). Same regex ideas underneath.

Activity

In-class activity

Open `day23-activity.R` in RStudio and work through the parts in order. (Work on the activity when it isn’t your group’s turn for **oral check-in #1**.)

- **Part 1:** detect patterns with `grepl()` / `str_detect()` + `filter()` on `BabyNames`.
- **Part 2:** build patterns from **character classes**, **quantifiers**, and **anchors** (including a “Caitlin”-style name and a phone-number regex).
- **Part 3 (challenge):** **extract** with capture groups, and **clean** messy strings with `gsub()`.

[Activity file](#)

Wrap-up & looking ahead

What we covered

- A regex describes a **pattern**; it does three jobs: **detect** / **replace** / **extract**
- . any char · [aeiou] class · [^aeiou] negation · | or
- Quantifiers ? * + {n} {n,m} {n,} (R: + = one-or-more, * = zero-or-more)
- Anchors ^ start, \$ end; **escape** specials with \
- Capture with (); pull pieces with `str_extract()` or `\\1` in `gsub()`
- Clean numbers/whitespace with `gsub()` (or `str_remove_all/str_squish`)

Upcoming

- **Fri Jun 19: No class — Juneteenth**
- **Quiz 5** rescheduled to **Mon Jun 22**
- Keep building your **project**; oral check-in #1 today

Reach out

Stuck? Office hours, or email me (cgc5478@psu.edu) or Xinyue (xpw5228@psu.edu).

Reference

[Syllabus](#) · [AI policy](#) · [Schedule](#) · [DC Ch. 17](#) · regex101.com